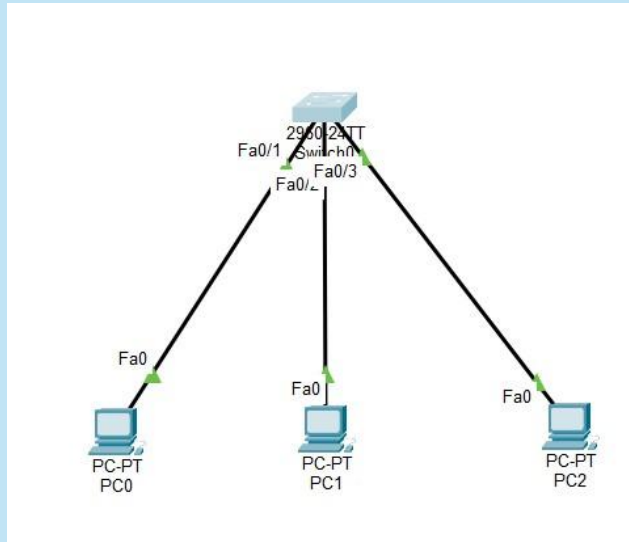


Inter vlan routing theory

1. In Cisco Packet Tracer, create three VLANs named Foodies, Shoppers, and Streamers to represent users of Zomato, Flipkart, and Spotify, and assign at least one PC to each VLAN.



Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#vlan 10
Switch(config-vlan)#name foodies
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name shoppers
Switch(config-vlan)#exit
Switch(config)#name streamers
^
% Invalid input detected at '^' marker.

Switch(config)#vlan 30
Switch(config-vlan)#name streamers
Switch(config-vlan)#exit
Switch(config)#int f0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#swi
Switch(config-if)#switchport access vlan 10
Switch(config-if)#exit
Switch(config)#int f0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#exit
Switch(config)#int f0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#exit
Switch(config)#show vlan brief
^
% Invalid input detected at '^' marker.

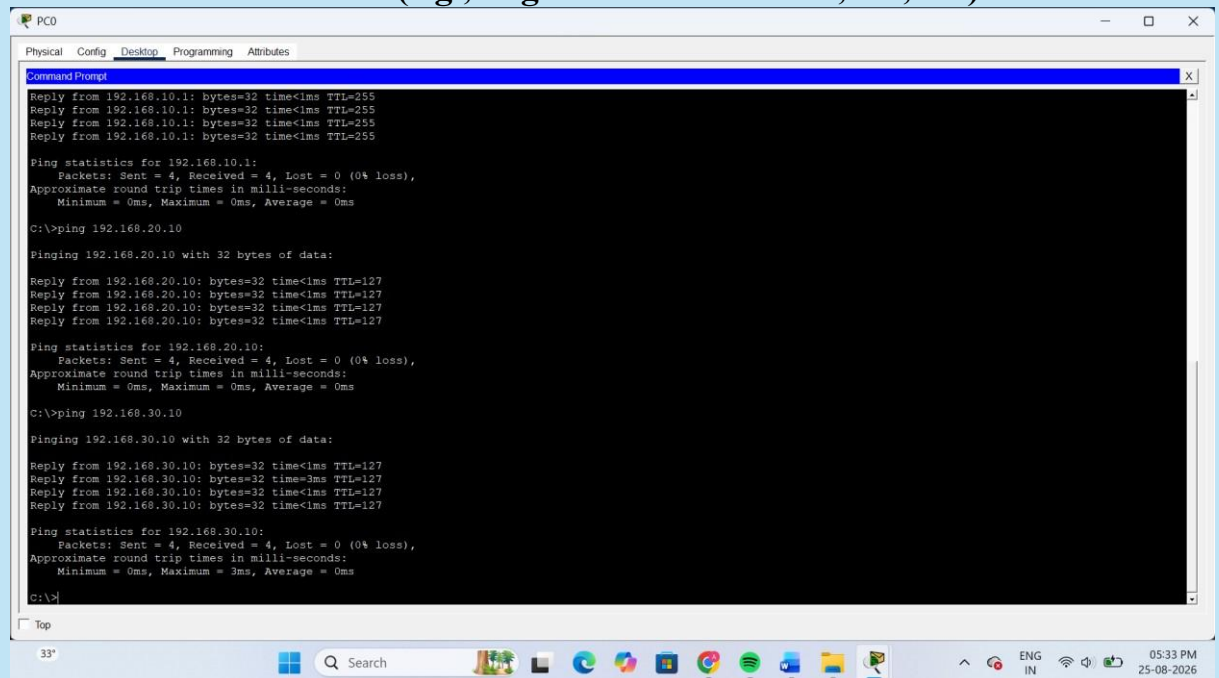
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10 foodies	active	Fa0/1
20 shoppers	active	Fa0/2
30 streamers	active	Fa0/3
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

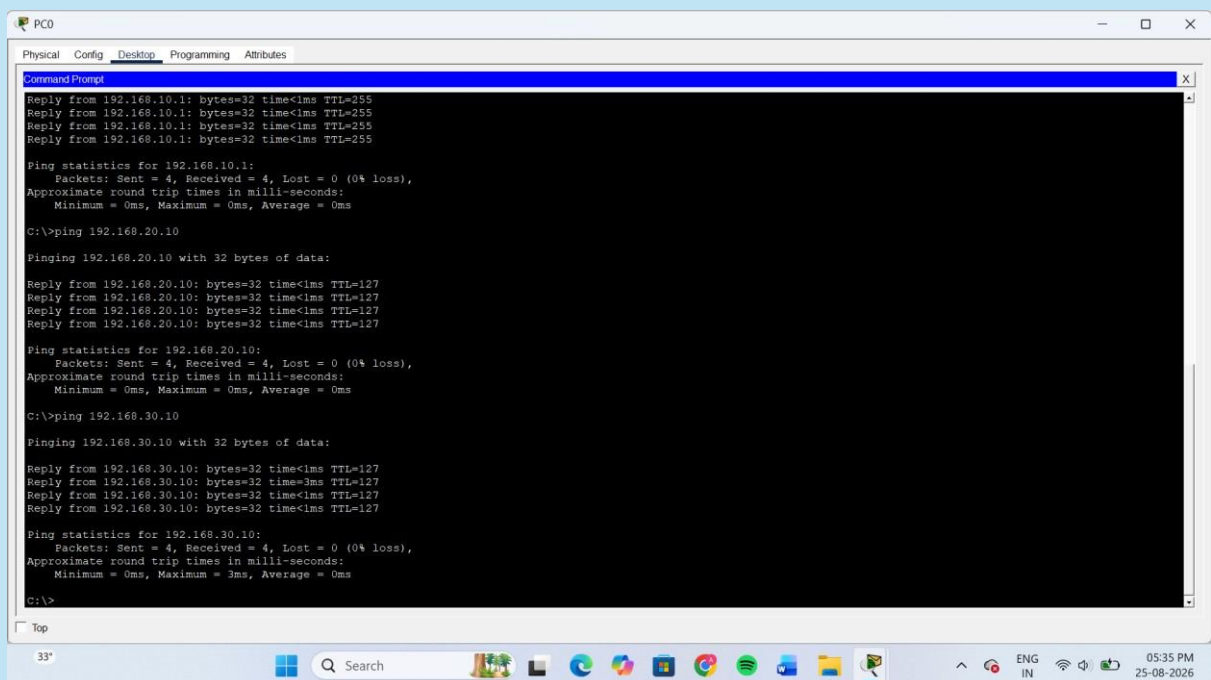
```
Switch#
```

2. Configure a router-on-a-stick setup in Packet Tracer so that PCs in the Foodies, Shoppers, and Streamers VLANs can all ping each other using sub-interfaces on the router (e.g., GigabitEthernet0/0.10, .20, .30). ○



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.20.10
Pinging 192.168.20.10 with 32 bytes of data:
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.30.10
Pinging 192.168.30.10 with 32 bytes of data:
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>
```

3. Assign IP addresses to each PC in your VLANs and set their default gateways to the corresponding router sub-interface IPs to enable interVLAN communication. ○



```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.20.10
Pinging 192.168.20.10 with 32 bytes of data:
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.30.10
Pinging 192.168.30.10 with 32 bytes of data:
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Reply from 192.168.30.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>
```

4. Simulate a scenario where users in the Shoppers VLAN (Flipkart) cannot ping users in the Streamers VLAN (Spotify). Identify and fix the misconfiguration causing the issue, then verify connectivity.
- Hint:** Check VLAN assignments, router sub-interface configurations, and trunk port settings.



```
Switch>
Switch>
Switch#enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface fastethernet 0/4
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 30
Switch(config-if)#no shutdown
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
Switch#
Switch#show vlan brief

VLAN Name                Status    Ports
----
1    default                active    Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                           Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                           Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                           Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                           Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                           Gi0/1, Gi0/2
10   Foodies                active    Fa0/2
20   Shoppers               active    Fa0/3
30   Streamers              active    Fa0/4
1002 fddi-default          active
1003 token-ring-default   active
1004 fddinet-default       active
1005 tnnnet-default       active
Switch#
```

```
Command Prompt
C:\>ping 192.168.30.10 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 3ms

C:\>ping 192.168.30.10

Pinging 192.168.30.10 with 32 bytes of data:
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127
Reply from 192.168.30.10: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.30.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

5. Compare the router-on-a-stick method you just configured with interVLAN routing using a Layer 3 switch: list two advantages and two disadvantages of each approach in a table.

○

Method	Advantages	Disadvantages
Routeron-aStick	1. Cost-effective: Uses one router interface with multiple subinterfaces, so no Layer 3 switch is required. 2. Easy to configure: Suitable for small networks and learning environments.	1. Limited performance: All inter-VLAN traffic passes through one router interface, which can become a bottleneck. 2. Single point of failure: If the router or trunk link fails, communication between all VLANs is affected.

Layer 3 Switch	1. High performance: Inter-VLAN routing is performed directly by the switch, providing faster communication. 2. Better scalability: Can handle many VLANs and high traffic volumes efficiently.	1. Higher cost: Layer 3 switches are generally more expensive than basic Layer 2 switches. 2. More complex: Configuration and troubleshooting can be more advanced than router-on-a-stick.
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